

Properties of Optimal Stimulus Spending

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Formula for optimal stimulus spending.

$$\frac{g/c - g/c^*}{g/c^*} = \frac{2 \overset{+}{\varepsilon} \overset{+}{m}}{1 + \underset{+}{2 \cdot \varepsilon} \underbrace{m^2}} \cdot \frac{u_0 - u^*}{u^*}$$

Role of ^{initial} unemployment gap, $u_0 - u^*$

Stimulus spending is larger when unemployment gap is larger

Role of elasticity of substitution, ε

$\varepsilon = 0 \rightarrow$ no stimulus spending ($g/c - g/c^*$)
(digging holes)

stimulus spending is $\uparrow$ in $\varepsilon \rightarrow$ higher substitutability, stimulus package is larger.

$$\varepsilon \rightarrow \infty \rightarrow \frac{g/c - g/c^*}{g/c^*} = \frac{2}{2m} \cdot \frac{u_0 - u^*}{u^*}$$

Role of unemployment multiplier.

$m = 0$: no stimulus spending ($g/c - g/c^*$)

small m: stimulus spending is $\uparrow$ in m

medium m: stimulus spending peaks

$$\begin{aligned} \frac{\partial}{\partial m} \left[\frac{2\varepsilon m}{1 + 2\varepsilon m^2} \right] &= 0 \Rightarrow \frac{2\varepsilon}{1} - \frac{2\varepsilon m \times 2 \times 2\varepsilon m}{[...]^2} = 0 \\ &\Rightarrow 2\varepsilon [1 + 2\varepsilon m^2] - 4\varepsilon^2 m^2 = 0 \\ &\Rightarrow 2\varepsilon + 2\varepsilon^2 m^2 - 4\varepsilon^2 m^2 = 0 \end{aligned}$$

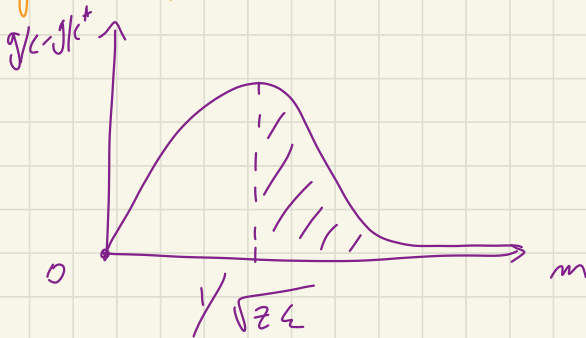
$$\Rightarrow 2z = [4z - 2z] z^2 m^2 = 2z z^2 m^2$$

$$\Rightarrow m^2 = \frac{2z}{2z z^2} = \frac{1}{z z}$$

$$\Rightarrow m^* = 1 / \sqrt{z z}$$

After $m > m^*$

stimulus spending $\downarrow$ w/ m



Raney (2013) $\rightarrow$ evidence on $m = -du/dg > 0$
 median estimate: $m \approx 0.5$